

Module-3

Laplace and Inverse Laplace transforms

* Laplace transforms :- If $f(t)$ is a real valued function defined for all $(t \geq 0)$ in the Laplace forms

of $f(t)$ is

$$L[f(t)] = \int_{t=0}^{t=\infty} e^{-st} f(t) dt$$

where $L[f(t)] = \bar{f}(s)$

$$\bar{f}(s) = \int_0^{\infty} e^{-st} f(t) dt$$

Formula

$f(t)$	$L[f(t)]$
a	$\frac{a}{s}$
e^{at}	$\frac{1}{s-a}$
e^{-at}	$\frac{1}{s+a}$
$\sin at$	$\frac{a}{s^2+a^2}$
$\cos at$	$\frac{s}{s^2+a^2}$

$\cosh at$

$$\frac{s}{s^2-a^2}$$

$\sinh at$

$$\frac{a}{s^2-a^2}$$

t^n

$$\frac{n!}{s^{n+1}} \quad n \text{ in (+ve) integer}$$

t^n

$$\frac{\Gamma(n+1)}{s^{n+1}}$$

$$\Gamma(n+1) = n!$$

$$\Gamma\frac{1}{2} = \frac{5}{2} \times \frac{3}{2} \times \frac{1}{2} \times \sqrt{\pi}$$

$$\Gamma\frac{3}{2} = \frac{7}{2} \times \frac{5}{2} \times \frac{3}{2} \times \frac{1}{2} \times \sqrt{\pi}$$

Problems:- (Laplace transforms)

Q1). $L[e^{2t}] = \frac{1}{s-2}$

• $L[e^{-4t}] = \frac{1}{s+4}$

• $L[\sin 3t] = \frac{3}{s^2+3^2} = \frac{3}{s^2+9}$

• $L[\cos 4t] = \frac{s}{s^2+4^2} = \frac{s}{s^2+16}$

• $L[\sinh 2t] = \frac{2}{s^2-2^2} = \frac{2}{s^2-4}$

• $L[\cosh 4t] = \frac{s}{s^2-4^2} = \frac{s}{s^2-16}$

$$\bullet L[t^5] = \frac{5!}{s^{5+1}} = \frac{5!}{s^6} \quad [5! = 5 \times 4 \times 3 \times 2 \times 1 = 120]$$

$$L[t^5] = \frac{120}{s^6}$$

$$\bullet L[t^{5/2}] = \frac{\Gamma(5/2+1)}{s^{5/2+1}}$$

$$L[t^{5/2}] = \frac{\Gamma(7/2)}{s^{7/2}}$$

$$L[t^{5/2}] = \frac{\frac{5}{2} \times \frac{3}{2} \times \frac{1}{2} \times \sqrt{\pi}}{s^{7/2}}$$

$$L[t^{5/2}] = \frac{15\sqrt{\pi}}{8s^{7/2}}$$

Q 2)

$$(i) L[\sin^2 2t]$$

$$\text{Solution:- } \sin^2 2t = \frac{1 - \cos 2(2t)}{2}$$

$$\sin^2 2t = \frac{1 - \cos 4t}{2}$$

$$L[\sin^2 2t] = \frac{1}{2} [L(1) - L(\cos 4t)]$$

$$= \frac{1}{2} \left[\frac{1}{s} - \frac{s}{s^2 + 16} \right]$$

$$L[\sin^2 2t] = \frac{1}{2} \left[\frac{1}{s} - \frac{s}{s^2 + 16} \right]$$

$$(ii) L[\sin 4t \cos 2t]$$

Solution:-

$$L[\sin 4t \cos 2t]$$

$$\sin A \cos B = \frac{1}{2} [\sin(A+B) + \sin(A-B)]$$

$$= \frac{1}{2} (\sin 6t + \sin 2t)$$

$$= \frac{1}{2} \left[\frac{6}{s^2 + 36} + \frac{2}{s^2 + 4} \right]$$

$$(iii) L[\sin 3t \cdot \sin t]$$

Solution:-

$$L[\sin 3t \cdot \sin t]$$

$$\sin A \sin B = \frac{1}{2} [\cos(A-B) - \cos(A+B)]$$

$$= \frac{1}{2} [\cos 2t - \cos 4t] \quad \left[\cos at = \frac{s}{s^2 + a^2} \right]$$

$$= \frac{1}{2} \left[\frac{s}{s^2 + 4} - \frac{s}{s^2 + 16} \right]$$

$$(iv) L[\cos 6t \cdot \cos 3t]$$

Solution:-

$$L[\cos 6t \cdot \cos 3t]$$

$$\cos A \cos B = \frac{1}{2} [\cos(A+B) + \cos(A-B)]$$

$$= \frac{1}{2} [\cos 9t + \cos 3t]$$

$$= \frac{1}{2} \left[\frac{s}{s^2 + 81} + \frac{s}{s^2 + 9} \right]$$

Q.3 Find $L[(3t+4)^3]$

Solution:- $L[(3t+4)^3]$

$$(a+b)^3 = a^3 + b^3 + 3ab(a+b)$$

$$L[(3t)^3 + (4)^3 + 3(3t)(4) + 3(3t)(4)]$$

$$= L[27t^3 + 64 + 108t^2 + 144t]$$

$$= 27L[t^3] + L[64] + 108L[t^2] + 144L[t]$$

$$= 27 \left(\frac{3!}{s^4} \right) + \frac{64}{s} + 108 \left(\frac{2!}{s^3} \right) + 144 \frac{1}{s^2}$$

$$= 27 \left(\frac{6}{s^4} \right) + \frac{64}{s} + 108 \left(\frac{2}{s^3} \right) + \frac{144}{s^2}$$

$$L[(3t+4)^3] = \frac{162}{s^4} + \frac{64}{s} + \frac{216}{s^3} + \frac{144}{s^2}$$

• $L(s^t)$

$$s^t = e^{(t \log s)}$$

$$L[s^t] = L[e^{(t \log s)}]$$

$$L[s^t] = \frac{1}{s - \log s}$$

Q.4 $L\left[\left(\sqrt{t} - \frac{1}{\sqrt{t}}\right)^3\right]$

Solution:- $L\left[\left(\sqrt{t} - \frac{1}{\sqrt{t}}\right)^3\right]$

$$(a-b)^3 = a^3 - b^3 - 3a^2b + 3ab^2$$

$$L\left[\left(\sqrt{t}\right)^3 - \left(\frac{1}{\sqrt{t}}\right)^3 - 3\left(\sqrt{t}\right)^2\left(\frac{1}{\sqrt{t}}\right) + 3\left(\sqrt{t}\right)\left(\frac{1}{\sqrt{t}}\right)^2\right]$$

$$= L\left[t^{3/2} - t^{-3/2} - 3t^{1/2} + 3t^{-1/2}\right]$$

$$= L\left[\left(\sqrt{t} - \frac{1}{\sqrt{t}}\right)^3\right] = L[t^{3/2}] - L[t^{-3/2}] - 3L[t^{1/2}] + 3L[t^{-1/2}]$$

$$\Rightarrow \frac{s^{-3/2+1}}{s^{-3/2+1}} - \frac{s^{-3/2+1}}{s^{-3/2+1}} - 3 \frac{s^{-1/2+1}}{s^{-1/2+1}} + 3 \frac{s^{-1/2+1}}{s^{-1/2+1}}$$

$$\Rightarrow \frac{s^{-1/2}}{s^{-1/2}} - \frac{s^{-1/2}}{s^{-1/2}} - 3 \frac{s^{1/2}}{s^{1/2}} + 3 \frac{s^{1/2}}{s^{1/2}}$$

$$\Rightarrow \frac{s^{-1/2}}{s^{-1/2}} = \frac{3}{2} \times \frac{1}{2} \times \sqrt{\pi}$$

$$\frac{s^{3/2}}{s^{3/2}} = \frac{1}{2} \times \sqrt{\pi}$$

$$\frac{s^{-3/2}}{s^{-3/2}} = \frac{1}{-3/2} \times \sqrt{\pi}$$

$$\frac{s^{-1/2}}{s^{-1/2}} = \frac{1}{-1/2} \times \sqrt{\pi}$$

$$\frac{s^{-5/2}}{s^{-5/2}} = \frac{1}{-5/2} \times \sqrt{\pi}$$

$$L\left[\left(\sqrt{s} - \frac{1}{\sqrt{s}}\right)^3\right] = \frac{\frac{3}{2} \times \frac{1}{2} \times \sqrt{\pi}}{s^{5/2}} - (-2\pi) - 3\left[\frac{\sqrt{\pi}}{s^{3/2}}\right] + \frac{3\sqrt{\pi}}{s^{1/2}}$$

$$L\left[\left(\sqrt{s} - \frac{1}{\sqrt{s}}\right)^3\right] = \frac{3\sqrt{\pi}}{4s^{5/2}} + 2\pi - \frac{3\sqrt{\pi}}{2s^{3/2}} + \frac{3\sqrt{\pi}}{s^{1/2}}$$

* Shifting Property

If $L[f(t)] = \bar{F}(s)$ then

$$L[e^{at}f(t)] = \bar{F}(s-a)$$

$$L[e^{-at}f(t)] = \bar{F}(s+a)$$

Problems:

① $L[e^{2t} \sin 4t]$

Solution:-

$L[e^{2t} \sin 4t]$ is of the form

$$L[e^{at}f(t)] = \bar{F}(s-a)$$

$$L[e^{2t} \sin 4t] = L[\sin 4t]_{s \rightarrow s-2}$$

$$= \left(\frac{4}{s^2 + 4^2}\right)_{s \rightarrow s-2}$$

$$= \frac{4}{(s-2)^2 + 16}$$

$$L[e^{2t} \sin 4t] = \frac{4}{s^2 - 4s + 4 + 16}$$

$$= \frac{4}{s^2 - 4s + 20}$$

$$L[e^{2t} \sin 4t] = \frac{4}{s^2 - 4s + 20}$$

② $L[e^{-3t} t^3]$

Solution:-

$L[e^{-3t} t^3]$ is of the form

$$L[e^{-at}f(t)] = \bar{F}(s+a)$$

$$L[e^{-3t} t^3] = L[t^3]_{s \rightarrow s+3}$$

$$= \left(\frac{3!}{s^4}\right)_{s \rightarrow s+3}$$

$$L[e^{-3t} t^3] = \frac{6}{(s+3)^4}$$

$$(iii) \mathcal{L}[e^{2t} \cos^2 4t]$$

It is of the form

$$\mathcal{L}[e^{at} f(t)] = \bar{f}(s-a)$$

$$\mathcal{L}[e^{2t} \cos^2 4t] = \mathcal{L}[\cos^2 4t]_{s \rightarrow s-2}$$

$$\cos^2 x = \frac{1 + \cos 2x}{2}$$

$$\mathcal{L}[e^{2t} \cos^2 4t] = \mathcal{L}\left[\frac{1 + \cos 8t}{2}\right]_{s \rightarrow s-2}$$

$$= \frac{1}{2} \{ \mathcal{L}(1) + \mathcal{L}(\cos 8t) \}_{s \rightarrow s-2}$$

$$= \frac{1}{2} \left[\frac{1}{s} + \frac{s}{s^2 + 8^2} \right]_{s \rightarrow s-2}$$

$$\mathcal{L}[e^{2t} \cos^2 4t] = \frac{1}{2} \left[\frac{1}{s-2} + \frac{s-2}{(s-2)^2 + 64} \right]$$

Q.P. Questions :-

$$(i) \mathcal{L}[e^{-3t} (2 \cos 5t - 3 \sin 5t)]$$

Solution :- $\mathcal{L}[e^{-3t} (2 \cos 5t - 3 \sin 5t)]$

$$= \mathcal{L}[2e^{-3t} \cos 5t] - \mathcal{L}[3e^{-3t} \sin 5t]$$

$$= 2 \mathcal{L}[e^{-3t} \cos 5t] - 3 \mathcal{L}[e^{-3t} \sin 5t]$$

$$= 2 \left\{ \mathcal{L}(\cos 5t) \right\}_{s \rightarrow s+3} - 3 \left\{ \mathcal{L}(\sin 5t) \right\}_{s \rightarrow s+3}$$

$$= 2 \left\{ \frac{s}{s^2 + 5^2} \right\}_{s \rightarrow s+3} - 3 \left\{ \frac{5}{s^2 + 5^2} \right\}_{s \rightarrow s+3}$$

$$\Rightarrow \frac{2(s+3)}{(s+3)^2 + 25} - \frac{15}{(s+3)^2 + 25}$$

$$= \frac{2s+6}{s^2 + 6s + 9 + 25} - \frac{15}{s^2 + 6s + 9 + 25}$$

$$= \frac{2s-9}{s^2 + 6s + 34}$$

$$\mathcal{L}[e^{-3t} (2 \cos 5t - 3 \sin 5t)] = \frac{2s-9}{s^2 + 6s + 34}$$

* $\frac{d}{ds} L[f(t)] = -f(s)$ then
 $L[t^n f(t)] = (-1)^n \frac{d^n}{ds^n} [L(f(t))]$
 where n is a (+ve) integer.

Q.
 (i) $L[t \cos at]$ (ii) $L[t^2 \sin t]$

Solution:-

$L[t \cos at]$ is of the form

$$L[t^n f(t)] = (-1)^n \frac{d^n}{ds^n} [L(f(t))]$$

$n=1$

$$L[t \cos at] = (-1) \frac{d}{ds} [L(\cos at)]$$

$$= -\frac{d}{ds} \left[\frac{s}{s^2+a^2} \right]$$

$$= - \left[\frac{(s^2+a^2)(1) - s(2s)}{(s^2+a^2)^2} \right]$$

$$L[t \cos at] = - \left[\frac{s^2+a^2-2s^2}{(s^2+a^2)^2} \right]$$

$$= - \left[\frac{-s^2+a^2}{(s^2+a^2)^2} \right]$$

$$L[t \cos at] = \frac{s^2-a^2}{(s^2+a^2)^2}$$

(ii) $L[t^2 \sin t]$

Solution:-

$L[t^2 \sin t]$ is of the form

$$L[t^n f(t)] = (-1)^n \frac{d^n}{ds^n} [L(f(t))]$$

$$L[t^2 \sin t] = (-1)^2 \frac{d^2}{ds^2} [L(\sin t)]$$

$$L[t^2 \sin t] = \frac{d^2}{ds^2} \left[\frac{1}{s^2+a^2} \right]$$

$$= \frac{d}{ds} \left[\frac{(s^2+a^2)(-2s) - 1(2s)}{(s^2+a^2)^2} \right]$$

$$= \frac{d}{ds} \left[\frac{-2s}{(s^2+a^2)^2} \right]$$

$$= - \left[\frac{(s^2+a^2)(-2) - 2s(2(s^2+a^2))}{(s^2+a^2)^4} \right]$$

$$= - \left[\frac{2(s^2+a^2) - 4s^2}{(s^2+a^2)^3} \right]$$

$$L[t^2 \sin t] = -2 \left[\frac{-3s^2 + a^2}{(s^2 + a^2)^3} \right]$$

$$L[t^2 \sin t] = 2 \left[\frac{3s^2 - a^2}{(s^2 + a^2)^3} \right]$$

$$(iii) L[t^5 e^{4t} \cosh 3t]$$

Solution:-

$$L[t^5 e^{4t} \cosh 3t]$$

$$\cosh 3t = \frac{e^{3t} + e^{-3t}}{2}$$

$$e^{4t} \cosh 3t = \frac{e^{4t}(e^{3t} + e^{-3t})}{2}$$

$$e^{4t} \cosh 3t = \frac{e^{7t} + e^t}{2}$$

$$t^5 e^{4t} \cosh 3t = \frac{t^5 (e^{7t} + e^t)}{2}$$

$$L[t^5 e^{4t} \cosh 3t] = \frac{1}{2} \{ L[t^5 e^{7t}] + L[t^5 e^t] \}$$

apply shift method

$$L[e^{at} f(t)] = \bar{f}(s-a)$$

$$= \frac{1}{2} \{ L[t^5]_{s \rightarrow s-7} + L[t^5]_{s \rightarrow s-1} \}$$

$$= \frac{1}{2} \left\{ \left(\frac{5!}{s^6} \right)_{s \rightarrow s-7} + \left(\frac{5!}{s^6} \right)_{s \rightarrow s-1} \right\}$$

$$= \frac{1}{2} \left\{ \frac{120}{(s-7)^6} + \frac{120}{(s-1)^6} \right\}$$

$$= \frac{120}{2} \left\{ \frac{1}{(s-7)^6} + \frac{1}{(s-1)^6} \right\}$$

$$= 60 \left\{ \frac{1}{(s-7)^6} + \frac{1}{(s-1)^6} \right\}$$

Q.P. Problem

$$(iv) L[t e^{-t} \sin 4t]$$

$$\underline{\text{Solution:-}} L[t e^{-t} \sin 4t]$$

$$L[e^{-t} t \sin 4t] = L[t \sin 4t]_{s \rightarrow s+1} \quad \text{[shifting property]}$$

$$L[t^n f(t)] = (-1)^n \frac{d^n}{ds^n} [L(f(t))]$$

$$L[e^{-t} t \sin 4t] = (-1) \frac{d}{ds} \{ L[\sin 4t] \}_{s \rightarrow s+1}$$

$$L[e^{-t} t \sin 4t] = -\frac{d}{ds} \left[\frac{4}{s^2 + 4^2} \right]_{s \rightarrow s+1}$$

$$= - \left[\frac{s^2 + a^2(2s) - 4(2s)}{(s^2 + 16)^2} \right]_{s \rightarrow s+1}$$

$$= \lim_{s \rightarrow s+1} \left[\frac{8s}{(s^2+16)^2} \right]$$

$$L[e^{-t} t \sin 4t] = \frac{8(s+1)}{(s+1)^2 + 16}$$

$$L[e^{-t} t \sin 4t] = \frac{8s+8}{(s^2+2s+17)^2}$$

if $L[f(t)] = \bar{f}(s)$ then

$$L\left[\frac{f(t)}{t}\right] = \int_s^\infty \bar{f}(s) ds$$

Find:-
(i) $L\left[\frac{\cos at - \cos bt}{t}\right]$

Solution:- $L\left[\frac{\cos at - \cos bt}{t}\right]$ is of the form

$$L\left[\frac{f(t)}{t}\right] = \int_s^\infty \bar{f}(s) ds$$

$$L\left[\frac{\cos at - \cos bt}{t}\right] = \int_s^\infty L(\cos at - \cos bt) ds$$

$$= \int_s^\infty \left(\frac{s}{s^2+a^2} - \frac{s}{s^2+b^2} \right) ds$$

Divide and multiply by (2)

$$= \frac{1}{2} \int_s^\infty \left(\frac{2s}{s^2+a^2} - \frac{2s}{s^2+b^2} \right) ds$$

$$= \frac{1}{2} \left[\log(s^2+a^2) - \log(s^2+b^2) \right]_s^\infty$$

$$= \frac{1}{2} \left\{ \log\left(\frac{s^2+a^2}{s^2+b^2}\right) \right\}_s^\infty$$

$$= \frac{1}{2} \left\{ \lim_{s \rightarrow \infty} \log\left(\frac{s^2+a^2}{s^2+b^2}\right) - \log\left(\frac{s^2+a^2}{s^2+b^2}\right) \right\}$$

$$= \frac{1}{2} \left[\log 1 - \log\left(\frac{s^2+a^2}{s^2+b^2}\right) \right]$$

$$= \frac{1}{2} \left[-\log\left(\frac{s^2+a^2}{s^2+b^2}\right) \right]$$

$$= \frac{1}{2} \left[\log\left(\frac{s^2+b^2}{s^2+a^2}\right) \right]$$

$$= \log \sqrt{\frac{s^2+b^2}{s^2+a^2}}$$

$$L\left[\frac{\cos at - \cos bt}{t}\right] = \log \sqrt{\frac{s^2+b^2}{s^2+a^2}}$$

(ii) $L\left[\frac{\sin^2 t}{t}\right]$ (iii) $L\left[\frac{1-\cos at}{t}\right]$

(iv) $L\left[\frac{2 \sin t \sin st}{t}\right]$

$$(ii) L\left[\frac{\sin^2 t}{t}\right]$$

Solution:- It is of the form

$$L\left[\frac{f(t)}{t}\right] = \int_s^\infty L[f(t)] ds$$

$$L\left[\frac{\sin^2 t}{t}\right] = \int_s^\infty L[\sin^2 t] ds$$

$$= \int_s^\infty L\left[1 - \frac{\cos 2t}{2}\right] ds$$

$$L\left[\frac{\sin^2 t}{t}\right] = \frac{1}{2} \int_s^\infty \{L(1) - L[\cos 2t]\} ds$$

$$= \frac{1}{2} \int_s^\infty \left(\frac{1}{s} - \frac{s}{s^2 + 2^2}\right) ds$$

$$= \frac{1}{2} \left\{ \int_s^\infty \frac{1}{s} ds - \int_s^\infty \frac{s}{s^2 + 2^2} ds \right\}$$

= multiply and divide by (2)

$$\frac{t}{dt} = \frac{s^2 + 2^2}{2s ds}$$

$$= \frac{1}{2} \left\{ \log s - \frac{1}{2} \log (s^2 + 2^2) \right\}_s^\infty$$

$$= \frac{1}{2} \left\{ \log s - \log \sqrt{s^2 + 2^2} \right\}_s^\infty$$

$$= \frac{1}{2} \left\{ \log \left(\frac{s}{\sqrt{s^2 + 2^2}} \right) \right\}_s^\infty$$

$$L\left[\frac{\sin^2 t}{t}\right] = \frac{1}{2} \left\{ \log 1 - \log \left(\frac{s}{\sqrt{s^2 + 2^2}} \right) \right\}$$

$$= \frac{1}{2} \left\{ 0 - \log \frac{s}{\sqrt{s^2 + 2^2}} \right\}$$

$$= \frac{1}{2} \left\{ \log \left(\frac{\sqrt{s^2 + 2^2}}{s} \right) \right\}$$

$$L\left[\frac{\sin^2 t}{t}\right] = \sqrt{\log \left(\frac{\sqrt{s^2 + 2^2}}{s} \right)}$$

$$(iii) L\left[\frac{1 - \cos at}{t}\right] \quad \text{Q.P}$$

Solution:- It is of the form

$$L\left[\frac{f(t)}{t}\right] = \int_s^\infty L[f(t)] ds$$

$$L\left[\frac{1 - \cos at}{t}\right] = \int_s^\infty L[1 - \cos at] ds$$

$$= \int_s^\infty \{L(1) - L[\cos at]\} ds$$

$$= \int_s^\infty \left(\frac{1}{s} - \frac{s}{s^2 + a^2}\right) ds$$

$$= \left\{ \int_s^\infty \frac{1}{s} ds - \int_s^\infty \frac{s}{s^2 + a^2} ds \right\}$$

$$= \left\{ \int_s^\infty \frac{1}{s} ds - \int_s^\infty \frac{s}{s^2+a^2} ds \right\}$$

$$= \left\{ \log s - \frac{1}{2} \log (s^2+a^2) \right\}_s^\infty$$

$$= \left\{ \log s - \log \sqrt{s^2+a^2} \right\}_s^\infty$$

$$= \left\{ \log \left(\frac{s}{\sqrt{s^2+a^2}} \right) \right\}_s^\infty$$

$$= \left\{ \log 1 - \log \left(\frac{s}{\sqrt{s^2+a^2}} \right) \right\}$$

$$= \left\{ 0 - \log \frac{s}{\sqrt{s^2+a^2}} \right\}$$

$$L \left[\frac{1 - \cos at}{x} \right] = \log \left(\frac{\sqrt{s^2+a^2}}{s} \right)$$

MQ.2]
(iv) $L \left[\frac{2 \sin t \cdot \sin st}{x} \right]$

Solution:- It is of the form

$$L \left[\frac{f(t)}{x} \right] = \int_s^\infty L[f(t)] ds$$

$$L \left[\frac{2 \sin t \sin st}{x} \right] = \int_s^\infty L[2 \sin t \sin st] ds$$

$$= 2 \int_s^\infty L[\sin t \sin st] ds$$

$$\therefore \sin A \sin B = \frac{1}{2} [\cos(A-B) - \cos(A+B)]$$

$$= 2 \int_s^\infty \frac{1}{2} L[\cos(1-s)t - \cos(1+s)t] ds$$

$$= \frac{2}{2} \int_s^\infty L[\cos 4t - \cos 6t] ds$$

$$= \int_s^\infty \left(\frac{s}{s^2+4^2} - \frac{s}{s^2+6^2} \right) ds$$

$$\Rightarrow \frac{1}{2} \int_s^\infty \left(\frac{2s}{s^2+16} - \frac{2s}{s^2+36} \right) ds$$

$$= \frac{1}{2} \left[\log(s^2+16) - \log(s^2+36) \right]_s^\infty$$

$$= \frac{1}{2} \left[\log \left(\frac{s^2+16}{s^2+36} \right) \right]_s^\infty$$

$$L \left[\frac{2 \sin t \sin st}{x} \right] = \sqrt{\log \left(\frac{s^2+16}{s^2+36} \right)}$$

* Unique Step function Unit

→ The unit step function $u(t-a)$ or Heaviside function $u(t-a)$ is defined as $u(t-a) = \begin{cases} 0 & t \leq a \\ 1 & t > a \end{cases}$

where a is a positive constant.

Formula:-

$$(i) L[u(t-a)] = \frac{e^{-as}}{s}$$

$$(ii) L[f(t-a)u(t-a)] = e^{-as} \bar{f}(s)$$

where $L[f(t)] = \bar{f}(s)$

* Laplace transform of discrete function in terms of unit step fn

$$f(t) = \begin{cases} f_1(t) & 0 < t < a \\ f_2(t) & t \geq a \end{cases}$$

$$f(t) = f_1(t) + (f_2(t) - f_1(t))u(t-a)$$

$$\text{If } f(t) = \begin{cases} f_1(t) & 0 < t \leq a \\ f_2(t) & a \leq t \leq b \\ f_3(t) & t > b \end{cases}$$

$$f(t) = f_1(t) + [f_2(t) - f_1(t)]u(t-a) + [f_3(t) - f_2(t)]u(t-b)$$

Problems

Express the following in terms of unit step function and find their Laplace.

$$(i) f(t) = \begin{cases} t^2 & 0 \leq t \leq 2 \\ 4t & t > 2 \end{cases}$$

Solution:- $f(t) = \begin{cases} t^2 & 0 \leq t \leq 2 & f_1(t) \\ 4t & t > 2 & f_2(t) \end{cases}$

$$f(t) = t^2 + (4t - t^2)u(t-2)$$

$$L[f(t)] = L[t^2] + L[(4t - t^2)u(t-2)]$$

$$L[f(t)] = \frac{2!}{s^3} + \textcircled{A} \quad \text{--- (1)}$$

$L[(4t - t^2)u(t-2)]$ is of the form

$$L[f(t-a)u(t-a)] = e^{-as} \bar{f}(s) \quad \alpha = 2$$

$$L[f(t-2)u(t-2)] = e^{-2s} \bar{f}(s) \quad \text{--- (2)}$$

To find $\bar{f}(s)$

$$f(t-2) = 4t - t^2$$

Replace t by $(t+2)$

$$f(t+2-2) = 4(t+2) - (t+2)^2$$

$$= 4t + 8 - (t^2 + 4t + 4)$$

$$= 4t + 8 - t^2 - 4t - 4$$

$$f(t) = -t^2 + 4$$

$$L[f(t)] = -L[t^2] + L[4]$$

$$\bar{f}(s) = \frac{-2!}{s^3} + \frac{4}{s}$$

$$\boxed{\bar{f}(s) = \frac{-2}{s^3} + \frac{4}{s}}$$

from ③

$$L[f(t-2)u(t-2)] = e^{-2s} \left(\frac{-2}{s^3} + \frac{4}{s} \right) \quad \text{--- ③}$$

Substitute ③ in ①

$$\boxed{L[f(t)] = \frac{2!}{s^3} + e^{-2s} \left(\frac{-2}{s^3} + \frac{4}{s} \right)}$$

$$(ii) f(t) = \begin{cases} t, & 0 < t \leq 4 \\ 5-t, & t > 4 \end{cases}$$

Solution:- $f(t) = \begin{cases} t, & 0 < t \leq 4 \\ 5-t, & t > 4 \end{cases}$

$$f(t) = t + (5-t)u(t-4)$$

$$L[f(t)] = L[t] + L[(5-t)u(t-4)]$$

$$L[f(t)] = \frac{1}{s^2} + \text{--- ①}$$

$$A = L[(5-t)u(t-4)]$$

is of the form

$$L[f(t-a)u(t-a)] = e^{-as} \bar{f}(s)$$

$$a=4$$

$$L[f(t-4)u(t-4)] = e^{-4s} \bar{f}(s) \quad \text{--- ②}$$

$$f(t-4) = 5-t$$

replace t by $t+4$

$$f(t+4) = 5-(t+4)$$

$$f(t+4) = 1-t$$

$$f(t) = L[1] - L[t]$$

$$f(s) = \frac{1}{s} - \frac{1}{s^2}$$

from ②

$$L[f(t-4)u(t-4)] = e^{-4s} \left(\frac{1}{s} - \frac{1}{s^2} \right) \quad \text{--- ③}$$

Substitute in ①

$$L[f(t)] = \frac{1}{s^2} + e^{-4s} \left(\frac{1}{s} - \frac{1}{s^2} \right)$$

$$(iii) f(t) = \begin{cases} 1 & 0 < t \leq 1 \\ t & 1 < t \leq 2 \\ t^2 & t > 2 \end{cases}$$

Solution:-

$$\begin{cases} 1 & 0 < t \leq 1 & f_1(t) \\ t & 1 < t \leq 2 & f_2(t) \\ t^2 & t > 2 & f_3(t) \end{cases}$$

$$f(t) = f_1(t) + (f_2(t) - f_1(t))u(t-a) + (f_3(t) - f_2(t))u(t-b)$$

$$a = 1, b = 2$$

$$f(t) = 1 + [(t-1)u(t-1) + (t^2-t)u(t-2)]$$

$$L[f(t)] = L[1] + L[(t-1)u(t-1)] + L[(t^2-t)u(t-2)]$$

$$L[f(t)] = \frac{1}{s} + \textcircled{A} + \textcircled{B} \quad \text{--- (1)}$$

$$\textcircled{A} = L[(t-1)u(t-1)] \text{ in the form of}$$

$$L[f(t-a)u(t-a)] = e^{-as} \bar{f}(s)$$

$$a = 1$$

$$L[f(t-a)u(t-a)] \Rightarrow e^{-s} \bar{f}(s) \quad \text{--- (2)}$$

To find $\bar{f}(s)$

$$f(t-1) = t-1$$

Replace t by $(t+1)$

$$f(t) = t+1-1$$

$$f(t) = t$$

$$L[f(t)] = L[t]$$

$$f(s) = \frac{1}{s^2}$$

from (2)

$$L[f(t-a)u(t-a)] = e^{-s} \left(\frac{1}{s^2} \right) \quad \text{--- (3)}$$

$$\textcircled{B} = L[(t^2-t)u(t-2)]$$

$$L[f(t-b)u(t-b)] = e^{-bs} \bar{f}(s)$$

$$b = 2$$

$$L[(t^2-t)u(t-2)] = e^{-2s} \bar{g}(s) \quad \text{--- (4)}$$

To find $\bar{g}(s)$

$$g(t-2) = t^2-t$$

Replace t by $(t+2)$

$$g(t+2-2) = (t+2)^2 - (t+2)$$

$$= t^2 + 4t + 4 - t - 2$$

$$g(t) = t^2 + 3t + 2$$

$$L[g(t)] = L[t^2] + L[3t] + L[2]$$

$$g(s) = \frac{2!}{s^3} + 3 \frac{1!}{s^2} + \frac{2}{s}$$

Sub $g(s)$ in (4)

$$L[(t^2-t)u(t-2)] = e^{-2s} \left(\frac{2}{s^3} + \frac{3}{s^2} + \frac{2}{s} \right) \quad \text{--- (5)}$$

Sub (3) & (5) in (1)

$$L[f(t)] = \frac{1}{s} + e^{-s} \frac{1}{s^2} + e^{-2s} \left(\frac{2}{s^3} + \frac{3}{s^2} + \frac{2}{s} \right)$$

H.W

$$1) f(t) = \begin{cases} \cos t & 0 < t < \pi \\ \sin t & t > \pi \end{cases} \quad \text{--- (3)} \quad f(t) = \begin{cases} \sin t & 0 < t \leq \pi \\ \sin t & \pi < t \leq 2\pi \\ \sin 3t & t > 2\pi \end{cases}$$

$$2) f(t) = \begin{cases} \pi - t & 0 \leq t < \pi \\ \sin t & t > \pi \end{cases}$$

(iv) Q.P.

$$f(t) = \begin{cases} \cos t & 0 < t \leq \pi \\ \cos 2t & \pi < t \leq 2\pi \\ \cos 3t & t > 2\pi \end{cases}$$

$$f(t) = f_1(t) + \{f_2(t) - f_1(t)\}u(t-a) + \{f_3(t) - f_2(t)\}u(t-b)$$

$$a = \pi$$

$$b = 2\pi$$

$$f(t) = \cos t + [\cos 2t - \cos t]u(t-\pi) + [\cos 3t - \cos 2t]u(t-2\pi)$$

$$L[f(t)] = L[\cos t] + L[\cos 2t - \cos t]u(t-\pi) + L[\cos 3t - \cos 2t]u(t-2\pi)$$

$$L[f(t)] = \frac{s}{s^2+1} + \text{(A)} + \text{(B)} \quad \text{--- (1)}$$

$A = L[(\cos 2t - \cos t)u(t-\pi)]$ is of the form

$$L[f(t)u(t-a)] = e^{-as} \bar{f}(s)$$

$$L[f(t-\pi)u(t-\pi)] = e^{-\pi s} \bar{f}(s) \quad \text{--- (2)}$$

To find $\bar{f}(s)$

$$f(t-\pi) = \cos 2t - \cos t$$

replace t by $t+\pi$

$$f(t+\pi-\pi) = \cos 2(t+\pi) - \cos(t+\pi)$$

$$f(t) = \cos 2(t+\pi) - (-\cos t)$$

$$f(t) = \cos 2t + \cos t$$

$$L[f(t)] = L[\cos 2t] + L[\cos t]$$

$$\bar{f}(s) = \frac{s}{s^2+4} + \frac{s}{s^2+1}$$

$$\text{(A)} \quad L[f(t-\pi)u(t-\pi)] = e^{-\pi s} \left(\frac{s}{s^2+4} + \frac{s}{s^2+1} \right) \quad \text{--- (3)}$$

(B) $L[\cos 3t - \cos 2t]u(t-2\pi)$ is of the form

$$L[g(t-b)u(t-b)] = e^{-bs} \bar{g}(s)$$

$$L[g(t-2\pi)u(t-2\pi)] = e^{-2\pi s} \bar{g}(s) \quad \text{--- (4)}$$

To find $\bar{g}(s)$

$$g(t - 2\pi) = \cos 3t - \cos t$$

Replace t by $(t + 2\pi)$

$$g(t) = \cos 3(t + 2\pi) - \cos(t + 2\pi)$$

$$g(t) = \cos(3t + 6\pi) - \cos(t + 2\pi)$$

$$g(t) = \cos 3t - \cos t$$

$$L[g(t)] = L[\cos 3t - \cos t]$$

~~g(t)~~

$$\bar{g}(s) = \frac{s}{s^2 + 9} - \frac{s}{s^2 + 4}$$

Substitute $\bar{g}(s)$ in (7)

$$L[(\cos 3t - \cos t)u(t - 2\pi)] = e^{-2\pi s} \left(\frac{s}{s^2 + 9} - \frac{s}{s^2 + 4} \right) \quad \text{--- (5)}$$

Substitute (5) & (6) in (1)

$$L[f(t)] = \frac{s}{s^2 + 1} + e^{-\pi s} \left(\frac{s}{s^2 + 4} + \frac{s}{s^2 + 1} \right) + e^{-2\pi s} \left(\frac{s}{s^2 + 9} - \frac{s}{s^2 + 4} \right)$$

H.W

$$(1) f(t) = \begin{cases} \cos t & 0 < t < \pi \\ \sin t & t > \pi \end{cases}$$

$f_1(t)$
 $f_2(t)$

Solution:- It is of the form

$$f(t) = f_1(t) + \{f_2(t) - f_1(t)\}u(t - a)$$

$$a = \pi$$

$$f(t) = \cos t + (\sin t - \cos t)u(t - \pi)$$

$$L[f(t)] = L[\cos t] + L\{(\sin t - \cos t)u(t - \pi)\}$$

$$L[f(t)] = \frac{s}{s^2 + 1} + \text{--- (A) --- (1)}$$

$$L[(\sin t - \cos t)u(t - \pi)] \text{ is of the form}$$

$$L[(\sin t - \cos t)u(t - \pi)]$$

$$L[f(t - a)u(t - a)] = e^{-as} \bar{f}(s)$$

$$a = \pi$$

$$L[f(t - \pi)u(t - \pi)] = e^{-\pi s} \bar{f}(s) \quad \text{--- (2)}$$

To find $\bar{f}(s)$

$$f(t - \pi) = \sin t - \cos t$$

Replace t by $(t + \pi)$

$$f(t) = \sin(t + \pi) - \cos(t + \pi)$$

$$f(t) = -\sin t - (-\cos t)$$

$$f(t) = \cos t - \sin t$$

$$L[f(t)] = L[\cos t] - L[\sin t]$$

$$\bar{f}(s) = \left(\frac{s}{s^2+1} \right) - \left(\frac{1}{s^2+1} \right)$$

$$\bar{f}(s) = \left(\frac{s}{s^2+1} \right) - \left(\frac{1}{s^2+1} \right)$$

Substitute in eqn (2)

$$L[(\sin t) - (\cos t) u(t-\pi)] = e^{-\pi s} \bar{f}(s)$$

$$L[(\sin t) - (\cos t) u(t-\pi)] = e^{-\pi s} \left(\frac{s}{s^2+1} \right) - \left(\frac{1}{s^2+1} \right) \quad \text{--- (3)}$$

Substitute eqn (3) in (1),

$$L[f(t)] = \left(\frac{s}{s^2+1} \right) + e^{-\pi s} \left[\left(\frac{s}{s^2+1} \right) - \left(\frac{1}{s^2+1} \right) \right]$$

$$(2) f(t) = \begin{cases} \pi - t & 0 \leq t < \pi \\ \sin t & t > \pi \end{cases}$$

Solution

$$f(t) = f_1(t) + \{f_2(t) - f_1(t)\} u(t-a)$$

$$f(t) = (\pi - t) + \{\sin t - (\pi - t)\} u(t-\pi)$$

$$L[f(t)] = L[\pi - t] + L[\sin t - (\pi - t) u(t-\pi)]$$

$$L[f(t)] = \frac{\pi}{s} - \frac{1}{s^2} + \text{--- (1)}$$

$$L[\sin t - (\pi - t) u(t-\pi)]$$

$$A \Rightarrow L[\sin t - \pi + t] u(t-\pi)$$

It is of the form

$$L[f(t-a) u(t-a)] = e^{-as} \bar{f}(s)$$

$$a = \pi$$

$$L[f(t-\pi) u(t-\pi)] = e^{-\pi s} \bar{f}(s) \quad \text{--- (2)}$$

To find $\bar{f}(s)$

$$f(t-\pi) u(t-\pi) = \sin t - \pi + t$$

Replace t by $(t+\pi)$

$$f(t) = \sin(t+\pi) - \pi + (t+\pi)$$

$$f(t) = -\sin t - \pi + t + \pi$$

$$f(t) = -\sin t + t$$

$$L[f(t)] = -L[\sin t] + L[t]$$

$$f(s) = -\frac{1}{s^2+1} + \frac{1}{s^2}$$

$$f(s) = \left(\frac{1}{s^2} \right) - \left(\frac{1}{s^2+1} \right)$$

Substitute in (2)

$$L[f(t-\pi) u(t-\pi)] = e^{-\pi s} \left(\frac{1}{s^2} \right) - \left(\frac{1}{s^2+1} \right) \quad \text{--- (3)}$$

Substitute eqn (3) in (1)

$$L[f(t)] = \frac{\pi}{s} - \frac{1}{s^2} + e^{-\pi s} \left(\frac{1}{s^2} \right) - \left(\frac{1}{s^2+1} \right)$$

$$\textcircled{3} f(t) = \begin{cases} \sin t & 0 < t \leq \pi \\ 2\sin t & \pi < t \leq 2\pi \\ \sin 3t & t > 2\pi \end{cases}$$

Solution:- $f(t) = f_1(t) + \{f_2(t) - f_1(t)\}u(t-a) + \{f_3(t) - f_2(t)\}u(t-b)$
 $a = \pi, b = 2\pi$

$$f(t) = \sin t + (\sin 2t - \sin t)u(t-\pi) + (\sin 3t - \sin 2t)u(t-2\pi)$$

$$L[f(t)] = L[\sin t] + L[(\sin 2t - \sin t)u(t-\pi)] + L[(\sin 3t - \sin 2t)u(t-2\pi)]$$

$$L[f(t)] = \frac{1}{s^2+1} + \textcircled{A} + \textcircled{B} \quad \text{--- (1)}$$

$$\textcircled{A} = L[(\sin 2t - \sin t)u(t-\pi)]$$

It is of the form

$$L[f(t-a)u(t-a)] = e^{-as} \bar{f}(s)$$

$$a = \pi$$

$$L[f(t-\pi)u(t-\pi)] = e^{-\pi s} \bar{f}(s) \quad \text{--- (2)}$$

To find $\bar{f}(s)$

$$f(t-\pi) = \sin 2t - \sin t$$

Replace t by $(t+\pi)$

$$f(t) = \sin 2(t+\pi) - \sin(t+\pi)$$

$$= \sin(2t+2\pi) - \sin(t+\pi)$$

$$= \sin 2t - \sin t$$

$$f(t) = \sin 2t + \sin t$$

$$L[f(t)] = L[\sin 2t] + L[\sin t]$$

$$f(s) = \left(\frac{2}{s^2+4}\right) + \left(\frac{1}{s^2+1}\right)$$

$\bar{f}(s)$ sub in (2)

$$L[f(t-\pi)u(t-\pi)] = e^{-\pi s} \left(\frac{2}{s^2+4}\right) + \left(\frac{1}{s^2+1}\right) \quad \text{--- (3)}$$

$$\textcircled{B} = L[(\sin 3t - \sin 2t)u(t-2\pi)]$$

It is of the form

$$L[g(t-b)u(t-b)] = e^{-bs} \bar{g}(s)$$

$$b = 2\pi$$

$$L[g(t-2\pi)u(t-2\pi)] = e^{-2\pi s} \bar{g}(s) \quad \text{--- (4)}$$

To find $\bar{g}(s)$

$$g(t) = \sin 3t - \sin 2t$$

$$g(t-2\pi) = \sin 3(t-2\pi) - \sin 2(t-2\pi)$$

Replace t by $(t+2\pi)$

$$g(t) = \sin 3(t+2\pi) - \sin 2(t+2\pi)$$

$$g(t) = \sin 3t - \sin 2t$$

$$L[g(t)] = L[\sin 3t] - L[\sin 2t]$$

$$\bar{g}(s) = \frac{3}{s^2+9} - \frac{2}{s^2+4}$$

$\bar{g}(s)$ sub in ④

$$L[g(t-2\pi)u(t-2\pi)] = e^{-2\pi s} \left(\frac{3}{s^2+9} - \frac{2}{s^2+4} \right) \text{--- ⑤}$$

Sub ③ & ⑤ in ①

$$L[f(t)] = \frac{1}{s^2+1} + e^{-\pi s} \left(\frac{3}{s^2+9} \right) + \left(\frac{1}{s^2+1} \right) + e^{-2\pi s} \left(\frac{3}{s^2+9} - \frac{2}{s^2+4} \right)$$

Practice

$$f(t) = \begin{cases} \cos t & 0 \leq t \leq \pi \\ \sin t & \pi < t < 2\pi \\ 0 & t > 2\pi \end{cases}$$

Solution $\Rightarrow$

$$f(t) = f_1(t) + \{f_2(t) - f_1(t)\}u(t-\pi) + \{f_3(t) - f_2(t)\}u(t-2\pi)$$

$$f(t) = \cos t + \{1 - \cos t\}u(t-\pi) + \{\sin t - 1\}u(t-2\pi)$$

$$L[f(t)] = L[\cos t] + L[(1 - \cos t)u(t-\pi)] + L[(\sin t - 1)u(t-2\pi)]$$

$$L[f(t)] = \frac{s}{s^2+1} + \textcircled{A} + \textcircled{B} \text{--- ①}$$

$$\textcircled{A} = L[(1 - \cos t)u(t-\pi)]$$

It is of the form

$$L[f(t-a)u(t-a)] = e^{-as} \bar{f}(s)$$

$$a = \pi$$

$$L[f(t-\pi)u(t-\pi)] = e^{-\pi s} \bar{f}(s) \text{--- ②}$$

To find $\bar{f}(s)$

$$f(t-\pi) = 1 - \cos t$$

Replace t by $(t+\pi)$

$$f(t) = 1 - \cos(t+\pi)$$

$$L[f(t)] = L[1] + L[\cos t]$$

$$L[f(t)] = L[1] + L[\cos t]$$

$$\bar{f}(s) = \frac{1}{s} + \frac{s}{s^2+1}$$

$\bar{f}(s)$ sub in ②

$$L[f(t-\pi)u(t-\pi)] = e^{-\pi s} \left(\frac{1}{s} + \frac{s}{s^2+1} \right) \text{--- ③}$$

$$\textcircled{B} = L[(\sin t - 1)u(t-2\pi)]$$

q is of the form

$$L[g(t-b)u(t-b)] = e^{-bs} \bar{g}(s)$$

$$b = 2\pi$$

$$L[g(t-2\pi)u(t-2\pi)] = e^{-2\pi s} \bar{g}(s) \text{--- ④}$$

To find $\bar{g}(s)$

$$L[g(t-2\pi)u(t-2\pi)]$$

$$g(t-2\pi) = \sin t - 1$$

Replace t by $(t+2\pi)$

$$g(t) = \sin(t+2\pi) - 1$$

$$g(t) = \sin t - 1$$

$$L[g(t)] = L[\sin t] - L[1]$$

$$\bar{g}(s) = \frac{1}{s^2+1} - \frac{1}{s}$$

$\bar{f}(s)$ substitute in (9)

$$L[f(t-\pi)] = e^{-2\pi s} \left(\frac{1}{s^2+1} - \frac{1}{s^2} \right) \quad (5)$$

Substitute (3) & (4) in (1)

$$L[f(t)] = \frac{s}{s^2+1} + e^{-\pi s} \left(\frac{1}{s} + \frac{s}{s^2+1} \right) + e^{-2\pi s} \left(\frac{1}{s^2+1} - \frac{1}{s^2} \right)$$

$$= \frac{s}{s^2+1} + \frac{e^{-\pi s}}{s} + \frac{e^{-\pi s}s}{s^2+1} + \frac{e^{-2\pi s}}{s^2+1} - \frac{e^{-2\pi s}}{s^2}$$

$$= \frac{s}{s^2+1} + \frac{e^{-\pi s}}{s} + \frac{e^{-\pi s}s}{s^2+1} + \frac{e^{-2\pi s}}{s^2+1} - \frac{e^{-2\pi s}}{s^2}$$

(6) in (2)

$$f(t) = \left(\frac{s}{s^2+1} + \frac{e^{-\pi s}}{s} + \frac{e^{-\pi s}s}{s^2+1} + \frac{e^{-2\pi s}}{s^2+1} - \frac{e^{-2\pi s}}{s^2} \right) \quad (7)$$

$$= \frac{s}{s^2+1} + \frac{e^{-\pi s}}{s} + \frac{e^{-\pi s}s}{s^2+1} + \frac{e^{-2\pi s}}{s^2+1} - \frac{e^{-2\pi s}}{s^2}$$

$$= \frac{s}{s^2+1} + \frac{e^{-\pi s}}{s} + \frac{e^{-\pi s}s}{s^2+1} + \frac{e^{-2\pi s}}{s^2+1} - \frac{e^{-2\pi s}}{s^2}$$

* Laplace transform of periodic functions

" A function $f(t)$ is said to be a periodic function of period $T > 0$ if $f(t+nT) = f(t)$ where $n=1, 2, 3, \dots$

$$L[f(t)] = \frac{1}{1-e^{-sT}} \int_0^T e^{-st} f(t) dt \quad (8)$$

Problems

① Find Laplace transform of given

$$f(t) = \begin{cases} E & 0 < t < a \\ -E & a < t < 2a \end{cases} \quad \text{where } f(t+a) = f(t)$$

(Q.1)

$$\text{Q.2} \quad \text{If } f(t) = \begin{cases} t & 0 \leq t \leq a \\ 2a-t & a \leq t \leq 2a \end{cases} \quad \text{where } f(t+2a) = f(t)$$

Hence show that $L[f(t)] = \frac{1}{s^2} \tanh\left(\frac{as}{2}\right)$

③ A periodic function of period $\frac{2\pi}{\omega}$ is defined by

$$f(t) = \begin{cases} E \sin \omega t & 0 \leq t \leq \pi/\omega \\ 0 & \pi/\omega \leq t \leq 2\pi/\omega \end{cases}$$

Show that $L[f(t)] = \frac{E\omega}{(s^2+\omega^2)(1-e^{-2\pi s/\omega})}$

Solutions:-

① Periodic formula

$$L[f(t)] = \frac{1}{1-e^{-sT}} \int_0^T e^{-st} f(t) dt$$

$$L[f(t)] = \frac{1}{1-e^{-as}} \int_0^a e^{-st} f(t) dt$$

$$L[f(t)] = \frac{1}{1-e^{-as}} \left\{ \int_0^{a/2} e^{-st} E dt + \int_{a/2}^a e^{-st} (-E) dt \right\}$$

$$= \frac{1}{1-e^{-as}} \left\{ E \int_0^{a/2} e^{-st} dt - E \int_{a/2}^a e^{-st} dt \right\}$$

$$= \frac{1}{1-e^{-as}} \left\{ E \left[\frac{e^{-st}}{-s} \right]_0^{a/2} - E \left[\frac{e^{-st}}{-s} \right]_{a/2}^a \right\}$$

$$= \frac{1}{1-e^{-as}} \left\{ -\frac{E}{s} (e^{-as/2} - e^0) + \frac{E}{s} (e^{-as} - e^{-as/2}) \right\}$$

$$= \frac{E}{s(1-e^{-as})} \left\{ -e^{-as/2} + 1 + e^{-as} - e^{-as/2} \right\}$$

$$= \frac{E}{s(1-e^{-as})} \left\{ 1 - 2e^{-as/2} + e^{-as} \right\}$$

$$= \frac{E}{s(1-e^{-as})^2} \left\{ (1-e^{-as/2})^2 \right\}$$

$$= \frac{E}{s(1-e^{-as/2})(1+e^{-as/2})} (1-e^{-as/2})^2$$

$$L[f(x)] = \frac{E}{s} \left(\frac{1-e^{-as/2}}{1+e^{-as/2}} \right)$$

Multiply & divide by $e^{\frac{as}{4}}$

$$L[f(x)] = \frac{E}{s} \left\{ \frac{e^{as/4} - e^{-as/4}}{e^{as/4} + e^{-as/4}} \right\}$$

$$L[f(x)] = \frac{E}{s} \left\{ \frac{x \sinh\left(\frac{as}{4}\right)}{x \cosh\left(\frac{as}{4}\right)} \right\}$$

$$\boxed{L[f(x)] = \frac{E}{s} \tanh\left(\frac{as}{4}\right)}$$

$$(2) \quad f(x) = \begin{cases} x & 0 \leq x \leq a/2 \\ 2a-x & a/2 \leq x \leq a \end{cases}$$

Periodic formula

$$\boxed{T = 2a}$$

$$L[f(x)] = \frac{1}{1-e^{-sT}} \int_0^T e^{-st} f(t) dt$$

$$L[f(x)] = \frac{1}{1-e^{-2as}} \int_0^{2a} e^{-st} f(t) dt$$

$$= \frac{1}{1-e^{-2as}} \left\{ \int_0^{a/2} e^{-st} (t) dt + \int_{a/2}^a e^{-st} (2a-t) dt \right\}$$

$$= \frac{1}{1-e^{-2as}} \left\{ \left[\frac{t \cdot e^{-st}}{-s} - (1) \frac{e^{-st}}{s^2} \right]_0^{a/2} + \left[(2a-t) \frac{e^{-st}}{-s} - (-1) \frac{e^{-st}}{s^2} \right]_{a/2}^a \right\}$$

$$= \frac{1}{1-e^{-2as}} \left\{ \frac{-1}{s} (t e^{-st})_0^{a/2} - \frac{1}{s^2} (e^{-st})_0^{a/2} + \frac{-1}{s} [(2a-t) e^{-st}]_{a/2}^a + \frac{1}{s^2} (e^{-st})_{a/2}^a \right\}$$

$$= \frac{1}{1-e^{-2as}} \left\{ \frac{-1}{s} (a e^{-sa} - 0) - \frac{1}{s^2} (e^{-sa} - e^0) - \frac{1}{s} [(2a-a) e^{-sa} - (2a-a/2) e^{-sa/2}] + \frac{1}{s^2} [e^{-sa} - e^{-sa/2}] \right\}$$

$$= \frac{1}{1-e^{-2as}} \left\{ \left[\frac{a}{s} e^{-sa} - \frac{1}{s^2} e^{-sa} \right] + \frac{a}{s} e^{-sa} + \frac{1}{s^2} e^{-2as} - \frac{1}{s^2} e^{-as} \right\}$$

$$L[f(x)] = \frac{1}{1-e^{-2as}} \left\{ \frac{1}{s^2} - \frac{2}{s^2} e^{-as} + \frac{1}{s^2} e^{-2as} \right\}$$

$$L[f(x)] = \frac{1}{s^2(1-e^{-2as})} \left\{ 1 - 2e^{-as} + e^{-2as} \right\}$$

$$= \frac{1}{s^2(1-e^{-as})^2} \left\{ 1 - (e^{-as})^2 \right\}$$

$$L[f(t)] = \frac{1}{s^2(1+e^{-as})(1-e^{-as})}$$

$$L[f(t)] = \frac{1}{s^2} \frac{(1-e^{-as})}{1+e^{-as}}$$

multiply & divide by $e^{as/2}$

$$L[f(t)] = \frac{1}{s^2} \left(\frac{e^{-\frac{as}{2}} - e^{-\frac{3as}{2}}}{e^{\frac{as}{2}} + e^{-\frac{as}{2}}} \right)$$

$$L[f(t)] = \frac{1}{s^2} \left\{ \frac{\sinh(\frac{as}{2})}{\cosh(\frac{as}{2})} \right\}$$

$$L[f(t)] = \frac{1}{s^2} \tanh\left(\frac{as}{2}\right)$$

$$(iii) f(t) = \begin{cases} E \sin \omega t & 0 < t \leq \pi/\omega \\ 0 & \pi/\omega < t < 2\pi/\omega \end{cases}$$

$$f(t + \frac{2\pi}{\omega}) = f(t)$$

Solution:-

given $f(t)$ is a periodic f^n

of period $\frac{2\pi}{\omega}$

$$L[f(t)] = \frac{1}{1-e^{-sT}} \int_0^T e^{-st} f(t) dt$$

$$T = \frac{2\pi}{\omega}$$

$$L[f(t)] = \frac{1}{1-e^{-s(\frac{2\pi}{\omega})}} \int_0^{\frac{2\pi}{\omega}} e^{-st} f(t) dt$$

$$= \frac{1}{1-e^{-2\pi s/\omega}} \left\{ \int_0^{\pi/\omega} e^{-st} E \sin \omega t dt + \int_{\pi/\omega}^{\frac{2\pi}{\omega}} 0 dt \right\}$$

$$= \frac{E}{1-e^{-2\pi s/\omega}} \int_0^{\pi/\omega} e^{-st} \sin \omega t dt$$

$$= \int_0^{\pi/\omega} e^{ax} \sin bx dx = \frac{e^{ax}}{a^2+b^2} (a \sin bx + b \cos bx)$$

$$= \frac{E}{1-e^{-2\pi s/\omega}} \left\{ \frac{e^{-st}}{(-s)^2 + (\omega^2)} (-s \sin \omega t - \omega \cos \omega t) \right\}_0^{\pi/\omega}$$

$$= \frac{-E}{1-e^{-2\pi s/\omega}} \left[e^{-s\pi/\omega} \left(\sin \omega \left(\frac{\pi}{\omega} \right) + \omega \cos \omega \left(\frac{\pi}{\omega} \right) \right) - e^0 (\sin \omega 0 + \omega \cos \omega 0) \right]$$

$$= \frac{-E}{1-e^{-2\pi s/\omega}} \left[\right]$$

$$= \frac{-E}{(s^2 + \omega^2)(1-e^{-2\pi s/\omega})} \left\{ e^{-s\pi/\omega} (0 - \omega) - e^{0} (0 + \omega(1)) \right\}$$

$$= \frac{-E(-\omega)}{(s^2 + \omega^2)(1-e^{-2\pi s/\omega})} \left\{ e^{-s\pi/\omega} + 1 \right\}$$

$$= \frac{E\omega}{(s^2 + \omega^2)(1 - e^{-2\pi/\omega})} \left(1 + e^{-5\pi/\omega} \right)$$

$$L[f(x)] = \frac{E\omega}{(s^2 + \omega^2)(1 - e^{-2\pi/\omega})} \left(1 + e^{-5\pi/\omega} \right)$$

$$= \frac{E\omega}{(s^2 + \omega^2)(1 + e^{-\pi/\omega})(1 - e^{-\pi/\omega})}$$

$$L[f(x)] = \frac{E\omega}{(s^2 + \omega^2)(1 - e^{-\pi/\omega})}$$

$$\left\{ \left(\frac{1}{\omega} \right) \cos(\omega t) - \left(\frac{1}{\omega} \right) \cos(\omega t - \pi) \right\} = \frac{1}{\omega} \left[\cos(\omega t) + \cos(\omega t - \pi) \right]$$

$$\left\{ (1 - \cos(\omega t)) - (1 - \cos(\omega t - \pi)) \right\} = \left[\cos(\omega t) - \cos(\omega t - \pi) \right]$$

$$\left\{ 1 - \cos(\omega t) \right\} = \left[1 - \cos(\omega t) \right]$$

★ Inverse Laplace Transform

Function $\& L^{-1}[F(s)]$	Inverse Function
$\frac{1}{s}$	1
$\frac{1}{s-a}$	e^{at}
$\frac{1}{s+a}$	e^{-at}
$\frac{s}{s^2+a^2}$	$\cos at$
$\frac{a}{s^2+a^2}$	$\sin at$
$\frac{s}{s^2-a^2}$	$\cosh at$
$\frac{a}{s^2-a^2}$	$\sinh at$
$\frac{1}{s^{n+1}} \quad ; n > -1$	$\frac{t^{n+1}}{(n+1)!}$
$\frac{1}{s^{n+1}} \quad ; n = 1, 2, 3, \dots$	$\frac{t^n}{n!}$

Basic Problems:-

$$L^{-1}\left[\frac{1}{s-2}\right] = e^{2t}$$

$$L^{-1}\left[\frac{1}{s^2+1}\right] = \sin t$$

$$L^{-1}\left[\frac{s}{s^2+81}\right] = \cos 9t$$

$$L^{-1}\left[\frac{4}{s^2+4^2}\right] = \sin 4t$$

$$L^{-1}\left[\frac{1}{s^2+5^2}\right] = \frac{\sin 5t}{5}$$

$$L^{-1}\left[\frac{1}{s^3}\right] = \frac{t}{2!}$$

$$L^{-1}\left[\frac{1}{s^{5k}}\right] = \frac{t^{3k}}{\left[\frac{3}{2}+1\right]} = \frac{t^{3k}}{\frac{5}{2}} = \frac{2t^{3k}}{5}$$

* Inverse transform of completing square

$$L^{-1}[\bar{f}(s-1)] = e^{at} L^{-1}[\bar{f}(s)]$$

$$1) L^{-1}\left[\frac{1}{(s-2)^2}\right] = L^{-1}\left[\left(\frac{1}{s^2}\right)_{s-2 \rightarrow s}\right] = \frac{t}{1} + \frac{A}{6} = \frac{t}{1} + \frac{A}{6}$$

$$(1)(s-2) = (e^{2t}) L^{-1}\left[\frac{1}{s^2}\right] + (1)(s-2)(1)A = 1$$

$$(1)(s-2) = e^{2t} t$$

$$2) L^{-1}\left[\frac{1}{(s+4)^3}\right] = L^{-1}\left[\left(\frac{1}{s^3}\right)_{s+4 \rightarrow s}\right]$$

$$= e^{-4t} L^{-1}\left[\frac{1}{s^3}\right]$$

$$= e^{-4t} \frac{t^2}{2}$$

$$L^{-1}\left[\frac{1}{(s+4)^3}\right] = \frac{e^{-4t} \cdot t^2}{2}$$

$$\frac{1}{s} = 1$$

Inverse Transform by method of Partial fraction

The method of partial fractions is a technique of converting an algebraic function, where

$$\frac{\phi(s)}{\psi(s)} \quad [\text{where } \phi(s) \text{ is } < \psi(s)]$$

into a sum. Depending on the nature of the term in $\psi(s)$ we have to split into the sum of various terms with constants $a, b, c, \dots$ which can be determined.

Problems

① Find $L^{-1} \left[\frac{1}{s(s+1)(s+2)(s+3)} \right]$

Solution:- $F(s) = \frac{1}{s(s+1)(s+2)(s+3)}$

$$\frac{1}{s(s+1)(s+2)(s+3)} = \frac{A}{s} + \frac{B}{s+1} + \frac{C}{s+2} + \frac{D}{s+3} \quad \text{--- (1)}$$

$$1 = A(s+1)(s+2)(s+3) + B[s(s+2)(s+3)] + C[s(s+1)(s+3)] + D[s(s+1)(s+2)] \quad \text{--- (2)}$$

Put $s=0$ in (2)

$$1 = A(1)(2)(3) + B(0) + C(0) + D(0)$$

$$1 = 6A$$

$$\boxed{A = \frac{1}{6}}$$

put $s = -1$ in (2)

$$1 = A(0)(1)(2) + B(-1)(-1+2)(-1+3) + C(0) + D(0)$$

$$1 = A(0) + B(-1)(1)(2)$$

$$1 = -2B$$

$$\boxed{-\frac{1}{2} = B}$$

Put $s = -2$ in (2)

$$1 = A(0) + B(0) + C(0)(-2+1)(-2+3) + D(0)$$

$$1 = C(-2)(-1)(1)$$

$$\boxed{\frac{1}{2} = C}$$

Put $s = -3$ in (2)

$$1 = A(0) + B(0) + C(0) + D(-3)(-3+1)(-3+2)$$

$$1 = D(-3)(-2)(-1)$$

$$\boxed{-\frac{1}{6} = D}$$

Substitute A, B, C, D in (1) we get

$$\mathcal{L}^{-1}\left[\frac{1}{s(s+1)(s+2)(s+3)}\right] = \frac{1}{6}\mathcal{L}^{-1}\left[\frac{1}{s}\right] - \frac{1}{2}\mathcal{L}^{-1}\left[\frac{1}{s+1}\right] + \frac{1}{2}\mathcal{L}^{-1}\left[\frac{1}{s+2}\right] - \frac{1}{6}\mathcal{L}^{-1}\left[\frac{1}{s+3}\right]$$

$$\mathcal{L}^{-1}\left[\frac{1}{s(s+1)(s+2)(s+3)}\right] = \frac{1}{6}(1) - \frac{1}{2}e^{-t} + \frac{1}{2}e^{-2t} - \frac{1}{6}e^{-3t}$$

② Find $L^{-1} \left[\frac{4s+5}{(s+1)^2(s+2)} \right]$

Solution :-

$$\tilde{f}(s) = \frac{4s+5}{(s+1)^2(s+2)}$$

$$\frac{4s+5}{(s+1)^2(s+2)} = \frac{A}{s+1} + \frac{B}{(s+1)^2} + \frac{C}{s+2} \quad \text{--- (1)}$$

$$4s+5 = A(s+1)(s+2) + B(s+2) + C(s+1)^2 \quad \text{--- (2)}$$

Put $s = -1$ in (2)

$$4s+5 = A(0) + B(1) + C(0)$$

~~4s+5 =~~

$$4(-1)+5 = B$$

$$\boxed{B = 1}$$

Put $s = -2$ in (2)

$$4(-2)+5 = A(-2+1)(-2+2) + B(-2+2) + C(-2+1)^2$$

$$-3 = A(0) + B(0) + C(1)$$

$$\boxed{C = -3}$$

Put $s = 0$ in (2)

$$4(0)+5 = A(0+1)(0+2) + B(0+2) + C(0+1)^2$$

$$5 = A(1)(2) + 2B + C$$

$$5 = 2A + 2(1) - 3$$

$$5 - 2 + 3 = 2A$$

$$6 = 2A$$

$$\frac{6}{2} = A$$

$$3 = A$$

Substituted A, B, C in (1)

$$\frac{4s+5}{(s+1)^2(s+2)} = \frac{3}{s+1} + \frac{1}{(s+1)^2} - \frac{3}{s+2}$$

$$\frac{4s+5}{(s+1)^2(s+2)} = 3L^{-1}\left[\frac{1}{s+1}\right] + L^{-1}\left[\frac{1}{(s+1)^2}\right] - 3L^{-1}\left[\frac{1}{s+2}\right]$$

$$= 3e^{-t} + L^{-1}\left[\left(\frac{1}{s^2}\right)_{s+1 \rightarrow s}\right] - 3e^{-2t}$$

$$= 3e^{-t} + e^{-t}t - 3e^{-2t}$$

$$= 3e^{-t} + e^{-t}t - 3e^{-2t}$$

$$L^{-1}\left[\frac{4s+5}{(s+1)^2(s+2)}\right] = (3e^{-t} + e^{-t}t - 3e^{-2t})$$

$$\textcircled{3} L^{-1}\left[\frac{2s^2+5s-4}{s^3+s^2-2s}\right]$$

Solution:-

$$f(s) = \frac{2s^2+5s-4}{s^3+s^2-2s}$$

$$\frac{2s^2+5s-4}{s^3+s^2-2s} =$$

$$\frac{2s^2+5s-4}{s^2+s-2}$$

$$s^2+s-2 = (s-1)(s+2)$$

$$\frac{2s^2+5s-4}{s(s-1)(s+2)}$$

$$\left[\frac{2s^2+5s-4}{s(s-1)(s+2)}\right] = \frac{A}{s} + \frac{B}{s-1} + \frac{C}{s+2} \quad \text{--- (1)}$$

$$2s^2 + 5s - 4 = A(s-1)(s+2) + B(s(s+2)) + C(s(s-1)) \quad \text{--- (2)}$$

Put $s=0$ in (2)

$$-4 = A(-1)(2) + B(0) + C(0)$$

$$-4 = -2A$$

$$\boxed{A=2}$$

Put $s=1$ in (2)

$$2+5-4 = A(0)(3) + B(3) + C(0)$$

$$3 = A(0) + B(3) + C(0)$$

$$\boxed{B=1}$$

Put $s=-2$

$$2(-2)^2 + 5(-2) - 4 = A(-2-1)(-2+2) + B(-2(-2+2)) + C(-2(-2-1))$$

$$8 - 10 - 4 = A(0) + B(0) + 6C$$

$$-6 = 6C$$

$$\boxed{C=-1}$$

Substitute A, B, C in (1)

$$\mathcal{L}^{-1} \left[\frac{2s^2 + 5s - 4}{s(s-1)(s+2)} \right] = 2\mathcal{L}^{-1} \left[\frac{1}{s} \right] + \mathcal{L}^{-1} \left[\frac{1}{s-1} \right] - \mathcal{L}^{-1} \left[\frac{1}{s+2} \right]$$

$$= 2(1) + e^t - e^{-2t}$$

$$\boxed{\mathcal{L}^{-1} \left[\frac{2s^2 + 5s - 4}{s(s-1)(s+2)} \right] = 2 + e^t - e^{-2t} \quad \text{H.}}$$

* Solution of Differential Equation Using Laplace transform.

Formula:-

$$\bullet L[y'(x)] = s L[y(x)] - y(0)$$

$$\bullet L[y''(x)] = s^2 L[y(x)] - s y(0) - y'(0)$$

$$\bullet L[y'''(x)] = s^3 L[y(x)] - s^2 y(0) - s y'(0) - y''(0)$$

Problems

(i) $y''(x) - 2y'(x) + y(x) = e^{2x}$; $y(0) = 0$, $y'(0) = -1$
Solve by using Laplace transform.

Solution:-

$$y''(x) - 2y'(x) + y(x) = e^{2x}$$

$$L[y''(x)] - 2L[y'(x)] + L[y(x)] = L[e^{2x}]$$

$$\{s^2 L[y(x)] - s y(0) - y'(0)\} - 2\{s L[y(x)] - y(0)\} + L[y(x)] = \frac{1}{s-2}$$

given $y(0) = 0$, $y'(0) = -1$

$$\{s^2 L[y(x)] - s(0) - (-1)\} - 2\{s L[y(x)] - 0\} + L[y(x)] = \frac{1}{s-2}$$

$$(s^2 - 2s + 1) L[y(x)] + 1 = \frac{1}{s-2}$$

$$(\mathcal{S}^2 - 2\mathcal{S} + 1) L[y(t)] = \frac{1}{\mathcal{S} - 2} - 1$$

$$(\mathcal{S}^2 - 2\mathcal{S} + 1) L[y(t)] = \frac{1 - (\mathcal{S} - 2)}{\mathcal{S} - 2}$$

$$(\mathcal{S} - 1)^2 L[y(t)] = \frac{3 - \mathcal{S}}{(\mathcal{S} - 2)}$$

$$L[y(t)] = \left[\frac{3 - \mathcal{S}}{(\mathcal{S} - 2)(\mathcal{S} - 1)^2} \right] \text{--- (1)}$$

$$\frac{3 - \mathcal{S}}{(\mathcal{S} - 2)(\mathcal{S} - 1)^2} = \frac{A}{\mathcal{S} - 2} + \frac{B}{\mathcal{S} - 1} + \frac{C}{(\mathcal{S} - 1)^2} \text{--- (2)}$$

$$3 - \mathcal{S} = A(\mathcal{S} - 1)^2 + B(\mathcal{S} - 2)(\mathcal{S} - 1) + C(\mathcal{S} - 2) \text{--- (3)}$$

Put $\mathcal{S} = 1$

$$3 - 1 = A(1 - 1)^2 + B(1 - 2)(1 - 1) + C(1 - 2)$$

$$3 - 1 = A(0) + B(0) + C(-1)$$

$$\boxed{-2 = C}$$

Put $\mathcal{S} = 2$

$$1 = A(2 - 1)^2 + B(2 - 2)(2 - 1) + C(2 - 2)$$

$$\boxed{1 = A}$$

Put $\mathcal{S} = 0$

$$3 = A(-1)^2 + B(-2)(-1) + C(-2)$$

$$3 = A + 2B - 2C$$

$$3 = 1 + 2B - 2(-2)$$

$$3 = 1 + 4 + 2B$$

$$3 - 5 = 2B$$

$$-2 = 2B$$

$$\boxed{B = -1}$$

Substitute A, B, C in (2)

$$\mathcal{L}^{-1}\left[\frac{3-s}{(s-2)(s-1)^2}\right] = \mathcal{L}^{-1}\left[\frac{1}{s-2}\right] + \mathcal{L}^{-1}\left[\frac{-1}{s-1}\right] + \mathcal{L}^{-1}\left[\frac{-2}{(s-1)^2}\right]$$

$$= e^{2t} - e^t - 2\mathcal{L}^{-1}\left[\frac{1}{(s-1)^2}\right]$$

$$= e^{2t} - e^t - 2\mathcal{L}^{-1}\left[\frac{1}{(s^2)_{s-1 \rightarrow s}}\right]$$

$$= e^{2t} - e^t - 2e^t \mathcal{L}^{-1}\left[\frac{1}{(s^2)}\right]$$

$$\boxed{\mathcal{L}^{-1}\left[\frac{3-s}{(s-2)(s-1)^2}\right] = e^{2t} - e^t - 2e^t t}$$

(2) $y''(t) + 4y'(t) + 3y(t) = e^{-t}$ given $y(0) = y'(0) = 0$

Solution :-

$$y''(t) + 4y'(t) + 3y(t)$$

$$\mathcal{L}[y''(t)] + 4\mathcal{L}[y'(t)] + 3\mathcal{L}[y(t)] = \mathcal{L}[e^{-t}]$$

$$\{s^2\mathcal{L}[y(t)] - sy(0) - y'(0)\} + 4\{s\mathcal{L}[y(t)] - y(0)\} + 3\mathcal{L}[y(t)] = \frac{1}{s+1}$$

$$\{s^2 L[y(t)] - s(0) - 0\} + 4\{s L[y(t)] - 0\} + 3 L[y(t)] = \frac{1}{s+1}$$

$$(s^2 + 4s + 3) L[y(t)] = \frac{1}{s+1}$$

$$(s+3)(s+1) L[y(t)] = \frac{1}{s+1}$$

$$L[y(t)] = \frac{1}{(s+1)(s+3)(s+1)}$$

$$L[y(t)] = \frac{1}{(s+1)^2(s+3)}$$

$$y(t) = L^{-1} \left[\frac{1}{(s+1)^2(s+3)} \right] \quad \text{--- (1)}$$

$$\frac{1}{(s+1)^2(s+3)} = \frac{A}{s+3} + \frac{B}{s+1} + \frac{C}{(s+1)^2} \quad \text{--- (2)}$$

$$1 = A(s+1)^2 + B(s+1)(s+3) + C(s+3) \quad \text{--- (3)}$$

put $s = -1$ in (3)

$$1 = A(-1+1)^2 + B(-1+1)(-1+3) + C(-1+3)$$

$$1 = A(0) + B(0) + C(2)$$

$$1 = 2C$$

$$\boxed{C = \frac{1}{2}}$$

put $s = -3$ in (3)

$$1 = A(-3)^2 + B(0) + C(0)$$

$$1 = 9A$$

$$\boxed{A = \frac{1}{9}}$$

$$\text{put } s=0 \text{ in (3)}$$

$$1 = A(1) + B(3) + C(3)$$

$$1 = A + 3B + 3C$$

$$1 = \frac{1}{4} + 3B + 3\left(\frac{1}{2}\right)$$

$$1 = \frac{1}{4} + \frac{3 \times 2}{2 \times 2} + 3B$$

Multiply (3) with $\left(\frac{3}{2}\right)$

$$1 = \frac{1}{4} + 3B$$

$$1 - \frac{1}{4} = 3B$$

$$\frac{4-1}{4} = 3B$$

$$\frac{3}{4} = 3B$$

$$\boxed{B = \frac{1}{4}}$$

Substitute A, B, C in (2)

$$\frac{1}{(s+1)^2 + (3)^2} = \frac{1}{4} L^{-1}\left[\frac{1}{s+3}\right] - \frac{1}{4} L^{-1}\left[\frac{1}{s+1}\right] + \frac{1}{2} L^{-1}\left[\frac{1}{(s+1)^2}\right]$$

$$= \frac{1}{4} e^{-3t} - \frac{1}{4} e^{-t} + \frac{1}{2} L^{-1}\left[\frac{1}{(s^2)^{s+1 \rightarrow s}}\right]$$

$$= \frac{1}{4} e^{-3t} - \frac{1}{4} e^{-t} + \frac{1}{2} L^{-1}\left[\frac{1}{s^2}\right]$$

$$\boxed{y(t) = \frac{1}{4} e^{-3t} - \frac{1}{4} e^{-t} + \frac{1}{2} e^{-t} t}$$

(iii) $y''(x) + 4y'(x) + 4y(x) = e^{-x}$ given $y(0) = y'(0) = 0$

Solution:-

$$y''(x) + 4y'(x) + 4y(x) = e^{-x}$$

$$L[y''(x)] + 4L[y'(x)] + 4L[y(x)] = L[e^{-x}]$$

$$\{s^2 L[y(x)] - sy(0) - y'(0)\} + 4\{s L[y(x)] - y(0)\} + 4L[y(x)] = \frac{1}{s+1}$$

$$\{s^2 L[y(x)] - s(0) - 0\} + 4\{s L[y(x)] - 0\} + 4L[y(x)] = \frac{1}{s+1}$$

$$s^2 L[y(x)] + 4s L[y(x)] + 4L[y(x)] = \frac{1}{s+1}$$

$$s^2 + 4s + 4 + L[y(x)] = \frac{1}{s+1}$$

$$(s+2)^2 L[y(x)] = \frac{1}{(s+1)}$$

$$L[y(x)] = \frac{1}{(s+2)^2(s+1)}$$

$$L[y(x)] = \frac{1}{(s+1)(s+2)^2}$$

$$y(x) = L^{-1}\left[\frac{1}{(s+1)(s+2)^2}\right] \quad \text{--- (1)}$$

$$y(x) = \frac{A}{s+1} + \frac{B}{s+2} + \frac{C}{(s+2)^2}$$

$$\frac{1}{(s+1)(s+2)^2} = \frac{A}{(s+1)} + \frac{B}{(s+2)} + \frac{C}{(s+2)^2} \quad \text{--- (2)}$$

$$1 = A(s+2)^2 + B(s+1)(s+2) + C(s+1) \quad \text{--- (3)}$$

put $s = -2$

$$1 = A(-2+2) + B(-2+1)(-2+2) + C(-2+1)$$

$$1 = -C$$

$$\boxed{C = -1}$$

put $s = -1$

$$1 = A(-1+2) + B(-1+1)(-1+2) + C(-1+1)$$

$$1 = A + B(0) + C(0)$$

$$\boxed{1 = A}$$

put $s = 0$

$$1 = A(4) + 2B + C$$

$$1 = 4 + 2B + -1$$

$$1 = 3 + 2B$$

$$1 - 3 = 2B$$

$$-2 = 2B$$

$$\boxed{B = -1}$$

Substitute A, B, C in (2)

$$\frac{1}{(s+1)(s+2)^2} = \mathcal{L}^{-1} \left[\frac{1}{s+1} \right] - \mathcal{L}^{-1} \left[\frac{1}{s+2} \right] - \mathcal{L}^{-1} \left[\frac{1}{(s+2)^2} \right]$$

$$\frac{1}{(s+1)(s+2)^2} = e^{-t} - e^{-2t} - \mathcal{L}^{-1} \left[\frac{1}{(s+2)^2} \right]$$

$$\boxed{\frac{1}{(s+1)(s+2)^2} = e^{-t} - e^{-2t} - e^{-t} t}$$

④ $y''(x) + 3y'(x) + 2y(x) = 0$, $y(0) = 1$, $y'(0) = 0$

Solution:-

$$L[y''(x)] + 3L[y'(x)] + 2L[y(x)] = L[0]$$

$$\{s^2 L[y(x)] - s y(0) - y'(0)\} + 3\{s L[y(x)] - y(0)\} + 2L[y(x)] = 0$$

$$\{s^2 L[y(x)] - s(1) - 0\} + 3\{s L[y(x)] - 1\} + 2L[y(x)] = 0$$

$$s^2 + 3s + 2 L[y(x)] - 3 - s = 0$$

$$(s^2 + 3s + 2) L[y(x)] = 3 + s$$

$$L[y(x)] = \frac{3+s}{s^2+3s+2}$$

$$L[y(x)] = \frac{3+s}{(s+1)(s+2)}$$

$$y(x) = L^{-1} \left[\frac{s+3}{(s+1)(s+2)} \right] \quad \text{--- (1)}$$

$$\frac{s+3}{(s+1)(s+2)} = \frac{A}{s+1} + \frac{B}{s+2} \quad \text{--- (2)}$$

$$s+3 = A(s+2) + B(s+1)$$

Put $s = -1$

$$1 = A(0) - B$$

$$B = -1$$

put $s = -1$

$2 = A + 0$

$A = 2$

Substitute $A = 2, B = -1$ in (2)

$$\frac{s+3}{(s+1)(s+2)} = \frac{2}{s+1} - \frac{1}{s+2}$$

$$L^{-1}\left[\frac{s+3}{(s+1)(s+2)}\right] = 2L^{-1}\left[\frac{1}{s+1}\right] - L^{-1}\left[\frac{1}{s+2}\right]$$

$y(t) = 2e^{-t} - e^{-2t}$

* Convolution Theorem

Procedure:- 1) The given function express as the product of two functions say $\bar{f}(s)$ & $\bar{g}(s)$

2) We find $L^{-1}[\bar{f}(s)] = f(t)$ & $L^{-1}[\bar{g}(s)] = g(t)$

3) We applying convolution theorem

$$L^{-1}[\bar{f}(s) \cdot \bar{g}(s)] = \int_{u=0}^t f(u) g(t-u) du$$

4) Evaluate the integral to get the required result.

Q11 Using Convolution theorem obtain the inverse Laplace transform for the following functions

(i) $\frac{1}{s(s^2+a^2)}$ (ii) $\frac{s^2}{(s^2+a^2)^2}$ (iii) $\frac{1}{(s^2-1)(s^2+1)}$ (iv) $\frac{s}{(s^2+a^2)^2}$

$$L^{-1}\left[\frac{1}{s(s^2+a^2)}\right]$$

we know that

$$L^{-1}[f(s) \cdot g(s)] = \int_{u=0}^t f(u) g(t-u) du$$

$$\frac{1}{s(s^2+a^2)} = \frac{1}{s} \cdot \frac{1}{s^2+a^2}$$

$$f(s) = \frac{1}{s} \quad g(s) = \frac{1}{s^2+a^2}$$

$$L^{-1}[f(s)] = L^{-1}\left[\frac{1}{s}\right]; \quad L^{-1}[g(s)] = L^{-1}\left[\frac{1}{s^2+a^2}\right]$$

$$f(t) = 1$$

$$g(t) = \sin at$$

$$f(u) = 1$$

$$g(t-u) = \frac{\sin(at-au)}{a}$$

$$L^{-1}\left[\frac{1}{s(s^2+a^2)}\right] = \int_{u=0}^t 1 \cdot \frac{\sin(at-au)}{a} du$$

$$= \frac{1}{a} \int_{u=0}^t \sin(at-au) du$$

$$= \frac{1}{a} \left[\frac{\cos(at-au)}{a} \right]_0^t$$

$$\begin{aligned}
 \therefore \sin A \cos B &= \frac{1}{2} [\sin(A+B) + \sin(A-B)] \\
 &= \frac{1}{2} \int_0^t \frac{1}{2} \sin(au+at-au) + \sin(au-(at-au)) du \\
 &= \frac{1}{2a} \int_0^t (\sin at + \sin(2au-at)) du \\
 &= \frac{1}{2a} \left\{ \int_0^t \sin at du + \int_0^t \sin(2au-at) du \right\}
 \end{aligned}$$

$$\begin{aligned}
 \mathcal{L}^{-1} \left[\frac{s}{(s^2+a^2)^2} \right] &= \frac{1}{2a} \left\{ \sin at \int_0^t du + \int_0^t \sin(2au-at) du \right\} \\
 &= \frac{1}{2a} \left\{ \sin at (u)_0^t + \frac{1}{2} \left\{ -\cos(2au-at) \right\}_0^t \right\} \\
 &= \frac{1}{2a} \left\{ \sin at (t-0) - \frac{1}{2a} \left\{ \cos(2at-at) - \cos(2(0)-at) \right\} \right\} \\
 &= \frac{1}{2a} \left\{ \sin at (t) - \frac{\cos at}{2a} + \frac{\cos at}{2a} \right\}
 \end{aligned}$$

$$\boxed{\mathcal{L}^{-1} \left[\frac{s}{(s^2+a^2)^2} \right] = \frac{1}{2a} t \sin at}$$

(iii) $\frac{1}{(s-1)(s^2+1)}$

Solve:-

$$\mathcal{L}^{-1} \left[\frac{1}{(s-1)(s^2+1)} \right]$$

$$\mathcal{L}^{-1} [\bar{f}(s) \bar{g}(s)] = \int_{u=0}^t f(u) g(t-u) du$$

$$f(s) = \frac{1}{s-1} \quad ; \quad g(s) = \frac{1}{s^2+1}$$

$$L^{-1}\left[\frac{1}{s-1}\right] = L^{-1}\left[\frac{1}{s-1}\right] \quad ; \quad L^{-1}\left[\frac{1}{s^2+1}\right]$$

$$f(t) = e^t \quad ; \quad g(t) = \sin t$$

$$L^{-1}\left[\frac{1}{(s-1)(s^2+1)}\right] = \int_{u=0}^t e^u \sin(t-u) du$$

$$\int e^{ax} \sin bx dx = \frac{e^{ax}}{a^2+b^2} (a \sin bx + b \cos ax)$$

$$L^{-1}\left[\frac{1}{(s-1)(s^2+1)}\right] = \left[\frac{e^u}{-1^2+(-1)^2} \left(\sin(t-u) - (-1) \cos(t-u) \right) \right]_{u=0}^t$$

$$= \left[\frac{e^u}{2} \sin(t-u) + \cos(t-u) \right]_{u=0}^t$$

$$= \frac{1}{2} \left[e^u (\sin(t-u) + \cos(t-u)) \right]_0^t$$

$$= \frac{1}{2} \left[e^t (\sin(t-t) + \cos(t-t)) - e^0 (\sin(t-0) + \cos(t-0)) \right]$$

$$= \frac{1}{2} \left[e^t (\sin 0 + \cos 0) - (1 \sin t + \cos t) \right]$$

$$= \frac{1}{2} \left[e^t - \sin t + \cos t \right]$$

$$L^{-1}\left[\frac{1}{(s-1)(s^2+1)}\right] = \frac{1}{2} \left[e^t - \sin t + \cos t \right]$$

✓

M.Q.

$$\frac{1}{s^2(s^2+a^2)}$$

Solution:-

$$\mathcal{L}^{-1}\left[\frac{1}{s^2(s^2+a^2)}\right]$$

$$\mathcal{L}^{-1}[f(s) \cdot g(s)] = \int_{u=0}^t f(u) g(t-u) du$$

$$\frac{1}{s^2(s^2+a^2)}$$

$$f(s) = \frac{1}{s^2+a^2}, g(s) = \frac{1}{s^2}$$

$$\mathcal{L}^{-1}[f(s)] = \mathcal{L}^{-1}\left[\frac{1}{s^2+a^2}\right] \quad ; \quad \mathcal{L}^{-1}[g(s)] = \mathcal{L}^{-1}\left[\frac{1}{s^2}\right]$$

$$f(t) = \frac{\sin at}{a} \quad ; \quad g(t) = t$$

$$f(u) = \frac{\sin au}{a} \quad ; \quad g(t) = t-u$$

$$\mathcal{L}^{-1}[f(s) \cdot g(s)] = \int_{u=0}^t \frac{\sin au}{a} (t-u) du$$

$$= \frac{1}{a} \int_0^t (t-u) \sin au du$$

formula

$$\int u dv = uv - \int u'v_1 + u''v_2 \dots$$

$$u = t - a \quad \int du = \int \sin au$$

$$u' = -1 \quad v = -\frac{\cos au}{a}$$

$$\text{integration part } u_1' = -\frac{\sin au}{a^2}$$

$$= \frac{1}{a} \int \left((t-a) - \frac{\cos au}{a} - (-1) \left(-\frac{\sin au}{a^2} \right) \right) dt$$

$$= \frac{1}{a} \int \left[-\frac{1}{a} (t-a) \cos at - \frac{1}{a^2} (\sin at) \right] - (t-a) \cos a(0)$$

$$- \sin a(0) \Big]$$

$$= \frac{1}{a} \left\{ -\frac{1}{a} \left[(0) - (t) \right] \left[\frac{1}{a^2} \sin at \right] - (t-a) \cos a(0) \right\}$$

$$= \frac{1}{a} \left[\frac{t}{a} - \frac{1}{a^2} \sin at \right] \quad \text{I did different}$$

$$\boxed{L^{-1} [f(s) \cdot g(s)] = \frac{1}{a^2} - \frac{1}{a^3} \sin at}$$

Proof: Let \$f(s) = \frac{1}{s^2}\$ and \$g(s) = \frac{1}{s^3}\$.

Let \$W\$ be a subspace of \$V\$.

(i) \$W\$ is non-empty.

(ii) \$W\$ is closed under addition.

\$W \subseteq W + W \subseteq W \subseteq W\$.

(iii) \$W\$ is closed under scalar multiplication.

\$W \subseteq W \cdot W \subseteq W \subseteq W\$.

\$W \subseteq W \subseteq W\$.

Let \$W\$ be a subspace of \$V\$.

(i) \$W\$ is non-empty.

(ii) \$W\$ is closed under addition.

\$W \subseteq W \subseteq W\$.